Event-Related Alpha Desynchronization Brain Computer Interface Using Arduino UNO

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***Abstract*—This report details the hardware architecture and digital signal processing pipeline of a real-time Brain-Computer Interface (BCI) designed to detect Event-Related Desynchronization (ERD) in the Alpha band (8–12 Hz) in the Occipital lobe. By utilizing an isolated electromechanical actuation loop and dynamic statistical thresholding, the system successfully generalizes across various trials, maintaining a robust Signal-to-Noise Ratio while eliminating PWM-induced electrical artifacts.**

***Index Terms*—Brain-Computer Interface, Electroencephalography, Digital Signal Processing, Arduino, Alpha band.**

I. Introduction

THE Alpha rhythm (8–12 Hz) is highly responsive to visual stimuli and attention. The attenuation of this rhythm, known as the Berger effect or Event-Related Desynchronization (ERD), is a well-known phenomenon in biomedical literature. It was first discovered by the German neurologist and psychiatrist Hans Berger in 1929 [1]; in 1934, Edgar Douglas Adrian and Bryan Matthews conducted further studies on the Berger effect, locating it in the Occipital lobe of the scalp [2]; modern literature has described the ERD phenomenon in detail as a gain-control mechanism related to light adaptation [3], [4]. The primary scope of this project is to capture the biological ERD and translate it into a reliable control mechanism for external electromechanical hardware, allowing the user to control a small servomotor by monitoring the Alpha band spectral power. The system combines a compact design with audiovisual feedback, remaining fully operational in both user-operator and user-only setups. Furthermore, the entire BCI was successfully engineered with a total budget of roughly €100.

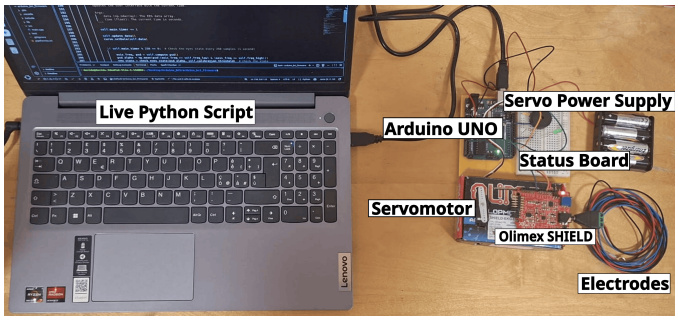


Fig. 1. Brain-Computer Interface setup.

II. Hardware Architecture & Noise Mitigation

To register the Occipital brain activity using electroencephalography (EEG), gold-plated (Au) electrodes have been used. Au was preferred to Ag/AgCl due to its durability and ease of maintenance, providing a suitable resolution for the experiment and a low impedance combined with conductive paste and a skin cleanser. The three electrodes are placed respectively on Oz (Channel +), Mastoid (Channel -), and Fz (Ground).

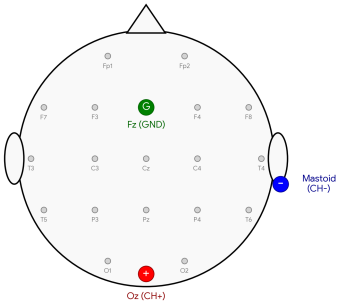


Fig. 2. Electrodes placement.

A significant challenge in closed-loop EEG systems is the acquisition of the signal and its amplification. To mitigate the noise of the EEG system, KONIX EEG Paste was applied to minimize impedance and secure the electrodes on the scalp. To amplify the signal from the usual $\sim \mu V$ range to the correct amplitude for the A/D converter, the electrodes are connected to the OLIMEX SHIELD EKG-EMG biosignal amplifier, with the following pipeline.

$$\text{Instr. Amp} \rightarrow \text{Besselworth 40Hz Lowpass Filter} \rightarrow \text{OAmp}$$

The OAmp was left with the default ($\sim$80) gain, bringing the $G_{\text{total}} = 10 \cdot 80 \cdot 3.46 = 2848$.

The SHIELD is powered by the Arduino board at 5 V and sends the digital data to the A0 port of the Arduino. The board also powers a green LED and a piezo speaker to provide visual and audio feedback to the user about the state of the machine. The EEG data is then sent in a 4 byte packet containing a start and a end sequence and the 10 bit signal in between. This configuration allows the computer program to correctly recognize the single data packet and synchronize with the board. A USB cable both powers the system and allows communication between the board and the computer.

The servomotor is powered separately with a 6 V battery to avoid PWM noise in the SHIELD. If the squared wave sent from the Arduino to power the servo is connected in parallel with the board, a small percentage of the signal enters the amplification pipeline, producing huge bumps that completely disrupt the signal.

III. BCI Pipeline

The entire BCI pipeline is shown below.

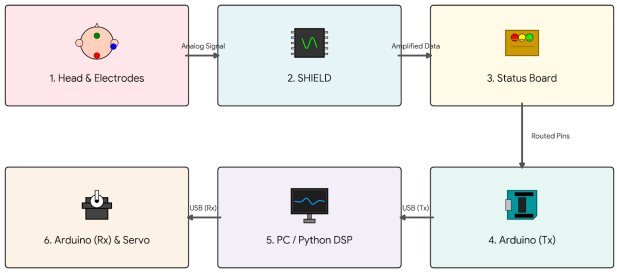


Fig. 3. The BCI Pipeline.

1) The electrodes placed on the head record the biopotential. The common mode signal is mitigated by the Fz reference electrode, and the Mastoid CH- is introduced as a differential input to subtract common muscle artifacts during the recording.
2) The SHIELD amplifies and filters the analog signal.
3) The board shows the BCI status. A green LED is illuminated during all entire data acquisition procedure.
4) The Arduino UNO converts the data to a digital 4-byte packet. Data is converted based on the 5 V scale according to the relation

$$\text{Data} = \frac{S}{V} \cdot 2^{10}$$

Where S is the analog signal and V is the power line of the board (5 V).
5) The PC reads the packet and analyzes the data. The processing is described in Section IV.
6) The PC feedback is sent back to the Arduino as an ASCII conversion of a char value, which is evaluated to correctly actuate the servo.

Meanwhile, a thread operation is conducted to track eye status and produce timestamps for eye opening/closing events. These data facilitate the offline analysis. The servo actuations are also saved with timestamps.

IV. Signal Processing

The continuous data stream is sampled at 250 Hz to provide a solid and precise signal. The digital signal processing is managed by the Python live script. The script is responsible for calibration, synchronization of the data collection process, data plotting, data filtering, spectral power estimation and response evaluation, and data storage. The choice of the coding language is justified by the useful libraries for signal processing and the ease of use for running live scripts.

A. Data collection

The script is set to accept data from a specified serial port; the packet received is then checked with the start and end sequence and the information content is then stored in a numerical array. The buffer records 5 minutes of data ($5 \cdot 60 \cdot F_s = 75 \cdot 10^3$ samples), providing a sufficient time window without the risk of RAM overflow. After exactly five minutes from the start of the experiment, the array is stored in a CSV file to allow for offline analysis, the script keeps running until stopped by the user.

B. Bandpass Filtering

A 4th-order Butterworth bandpass filter isolates the 8–12 Hz frequency band. Choosing to use Infinite Impulse Response (IIR) Filters allows low-latency filtering due to the low order, and a good mitigation of cutoff frequencies with a sufficient small ripple effect for the specific application.

C. Spectral Estimation and Windowing

Power Spectral Density (PSD) is calculated using Welch's method. The estimation is made on two second windows extracted from a 15 second frame during the calibration phase, and is then executed on every new second (250 samples) of data. Such time windows provide robust threshold estimation and dynamic power evaluation to correctly describe the power increase/decrease.

D. Multimodal Synchronization and Computer Vision

To establish an objective ground truth for the biological Event-Related Desynchronization, a computer vision pipeline was engineered to operate in parallel with the EEG acquisition. The **mediapipe Face Mesh framework** was utilized to extract facial landmarks in real-time, calculating the Eye Aspect Ratio (EAR) to determine the exact onset and offset of the user's eye closures.

Because the video feed operates at approximately 15 frames per second (FPS) while the EEG hardware samples at 250 Hz, running both processes on a single synchronous thread would bottleneck the signal acquisition loop, resulting in catastrophic data loss. To resolve this, a multithreaded architecture was implemented. The *EyeStateThread* operates entirely independently of the *EEGData* class.

To synchronize these disparate data streams, a common monotonic time base was established using the hardware's performance counter (`time.perf_counter()`). A global t_0 timestamp is generated at system initialization and shared across both threads. When the EAR algorithm detects a state change that exceeds the temporal debounce threshold, the exact differential timestamp is logged. This ensures that the low-frequency visual ground truth and the high-frequency electrophysiological data are aligned for offline validation, without risking the responsiveness of the servomotor.

V. Dynamic Calibration & Results

Because biological signals exhibit immense physiological variance across individuals, static amplitude thresholds frequently fail. The system provides high adaptability to different users by implementing an initial phase of calibration. The process is divided into 4 steps:

1) 15 second recording in active (eyes open) state. Estimation of low Alpha-band mean power m_{active}.
2) 15 second recording in rest (eyes closed) state. Estimation of high Alpha-band mean power m_{rest}.
3) Evaluation of spectral power levels. The script returns an error if the high level is below the lower level and prints a warning if the high level is less than 1.5 times the low level (which might be caused by bad electrode connection).
4) Estimation of threshold *T*.

A. Artifact Rejection & Outliers Robustness

To avoid including any movement that the user might do during the change from active (eyes open) and the rest (eyes closed) state, a transition time is added between the calibration steps: the first 3 seconds of both the windows are skipped and the PSD is evaluated over twelve overlapping windows from second 3 to second 15.

To mitigate servo twitching during the experiment—a behavior shown in early tests that was related to false positives/negatives and only lasted one window frame—the state evaluation system compares the new power window P_{new} to the last two windows evaluated $P_{current}$ and P_{past}. The algorithm logic is:

- if $P_{new} == P_{current}$ AND $P_{new} \neq P_{past}$: the servo changes its state.
- else: the servo does not change its state.

Plus, a 2 second refractory period is introduced after the activation/deactivation of the servo to mitigate the nonlinear alpha spike behavior: tests have shown a big initial spike followed by a lower energy period and then a plateau of high Alpha PSD.

Although this method introduces 3 seconds of latency to the actuation of the mechanism, it provides a much more precise response, mitigating false activations/deactivations.

B. Statistical Thresholding

The first version of the BCI operated with a threshold T_1 defined using a 3-sigma confidence interval based on the user's active (eyes open) Alpha power:

$$T_1 = m_{\text{active}} + 3 \cdot s_{\text{active}}$$

Where m_{active} and s_{active} are the sample's mean and standard deviation during the active phase (eyes open).

However, this method failed in users with low statistical difference between basal alpha activity and event-related desynchronization.

To develop a more general model, a coefficient α is proposed. A coefficient *C* is evaluated as the Median Absolute Deviation (MAD) of the active state of the subject normalized over the median of the PSD. The α is then determined by normalizing the coefficient over standard high and low *C* levels found empirically over numerous trials (in this case $C_L = 0.1$ and $C_H = 0.4$). These parameters can easily be tuned by the subject while testing the architecture, although the ultimate goal would be to generalize such parameters.

$$C = \frac{\text{median}(|PSD_{\text{active}} - M_{\text{active}}|)}{M_{\text{active}}}$$

$$\alpha = \frac{C - C_L}{C_H - C_L}$$

The final threshold is then determined by linear combination of a geometric mean (which highlights the logaritmic distribution of ERD) and the first method proposed. The median has been chosen as the standard method for the evaluation of the coefficient and the T_2 because of its mathematical robustness towards outliers—such as the spikes in power.

$$T_1 = m_{\text{active}} + 3 \cdot s_{\text{active}}$$

$$T_2 = \sqrt{M_{\text{active}} \cdot M_{\text{rest}}}$$

$$T = \alpha T_1 + (1 - \alpha) T_2$$

Being related to light adaptation, Alpha ERD is intrinsically a non-stationary process, so a dynamic threshold had to be implemented. The system effectively takes track of any valid input—so any input which satisfies the condition in section V-A—and uses the last 15 samples of high PSD and 15 samples of low PSD to evaluate a new threshold from scratch, adapting to the subject physiological variability.

C. Offline Validation

The new architecture has currently been tested over various trials on one subject. Future datasets are being produced.

Offline MATLAB analysis of continuous Alpha power revealed significant adaptability over different trials. To demonstrate adaptability, results over five trials are shown in Table I. The increase of latency from the first tests has been accepted as a trade-off for precision and robustness. The test showed an average precision (correct actuations over total actuations) of 81.5%, with the biggest flaw being false activations, which can be related to artifacts.

D. Latency and Event-Detection Accuracy

The system exhibits a median actuation latency of 3.91 seconds. This structural delay is not a computational flaw, but a deliberate mathematical aggregation of three factors: the biological habituation time required for the Alpha rhythm to fill the 2-second Welch periodogram, the 1.0-second tick quantization of the evaluation loop, and the temporal debounce mechanism (Section V-A) which enforces state stability.

However, when evaluating the architecture via Event-Detection Accuracy—defined as the successful actuation of the servomotor within the targeted biological window—the system achieved an evident True Positive catch. Across all validation trials, the hybrid median-MAD thresholding architecture successfully identified every intended physiological state change,

TABLE I
REAL-TIME BCI PERFORMANCE METRICS ACROSS FIVE VALIDATION TRIALS

Trial	P_{sig} ($\times 10^{-6}$)	P_{noise} ($\times 10^{-6}$)	**SNR (Lin)**	**SNR (dB)**	**Lat. (s)**
1	4.76	2.31	2.06	3.13	3.62
2	9.38	2.52	3.72	5.70	3.33
3	5.39	2.72	1.98	2.96	4.26
4	13.01	2.91	4.47	6.50	4.19
5	7.75	2.56	3.03	4.81	4.16
Avg	**8.06**	**2.61**	**3.05**	**4.62**	**3.91**

proving its efficacy as a highly reliable, although intentionally delayed, control mechanism.

The figure below shows the MATLAB offline analysis of the trial. Besides the figure, other useful statistics are evaluated and printed to the user. Pink windows are the closed eye frames, while gray windows are the open eye frames. The red and blue lines represent, respectively, servo actuation and deactivation.

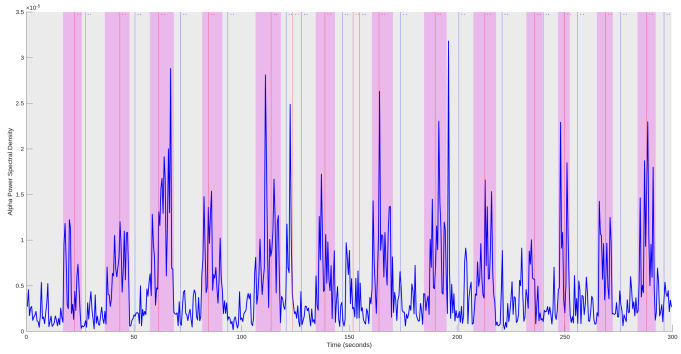


Fig. 4. MATLAB trial plot

VI. Discussion & Lessons Learned

Hardware-software co-design is critical in biomedical engineering. Even with correct electrode placement, DSP algorithms cannot work properly with a physically compromised signal. For this reason, securing the hardware grounding was a mandatory prerequisite for software filters and algorithms to function. Furthermore, the necessity of adaptive baselining was proven: static thresholds are unviable for general-purpose BCIs, providing good suitability for the majority of subjects.

Furthermore, while the hardware is robust, the biological Alpha ERD paradigm is intrinsically limited by user habituation to light intensity, making it suitable for simple binary triggers but highly impractical for continuous, complex prosthetic control.

VII. Conclusion

By combining strict hardware isolation with an adaptive, statistically grounded software threshold, this system provides a robust framework for real-time BCI control, generalizable across multiple users with minimal tuning.

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